

Claudespeak: Characterizing the Stylistic Fingerprint of a Frontier Language Model

Anonymous ACL submission

Abstract

Large language models are widely believed to have recognizable writing styles, but most popular claims (e.g. that models overuse the word *delve*) are derived from *ChatGPT* and applied indiscriminately to all models. We ask what, specifically, characterizes the prose of one frontier model—Anthropic’s Claude Opus 4.8—relative to human writing and to eight other models, holding topic constant via four parallel corpora that span informational QA, essays, real in-the-wild prompts, and ten professionally-authored task types. Combining interpretable stylometry, log-odds n -gram selection, a length-controlled regression, an interpretable classifier, and suffix-automaton (n -gram novelty) mining, we find a stable fingerprint that is *structural*, *rhythmic*, and *interactional* rather than a matter of stereotyped vocabulary. The fingerprint replicates with consistent sign across all four corpora, and Claude is identifiable from prose alone in every one (prose-only classifier AUC 0.79–0.95 across four independent contrasts), driven by markedly higher sentence-length variability, denser content-word usage, a strong em-dash habit, and—most diagnostically—a verbatim offer-to-continue closing move (“would you like me to go deeper into...”) that, in our corpus, appears dozens of times in Claude and *zero* times across humans and all other models. We further show that the stereotyped “delve” vocabulary is a GPT (and other-model) trait that Claude *avoids*: across seven models in an essay genre, Claude is the only model that never writes *delve* and uses *crucial* least. Finally, we audit corpus coverage and, finding the standard benchmarks narrow, broaden it with two diverse sources, releasing the fully provenanced corpus and pipeline. Sections on the effect of reasoning effort, Claude versions, and multi-turn dynamics are forthcoming.

1 Introduction

The claim that text “sounds like AI” is now commonplace, and a folk vocabulary of tells—*delve*, *tapestry*, *crucial*, *intricate*—circulates widely. Two problems undercut these claims as characterizations of any *particular* model. First, the evidence base is overwhelmingly about ChatGPT (Kobak et al., 2025), yet the conclusions are applied to “LLMs” generically. Second, most analyses do not control topic: a classifier that distinguishes model from human text may be learning the prompt distribution, not the writing style.

We study a single frontier model, Anthropic’s Claude Opus 4.8, and ask a deliberately narrow question: *holding the prompt fixed, what makes Claude’s prose identifiable relative to humans and to other models?* We build two topic-matched parallel corpora in which the same prompts are answered by Claude, by humans, and by up to six other models, and we attack the question with five complementary methods. Our contributions:

1. A parallel-corpus protocol and a fully provenanced dataset—four tracks, eight contrast models and two human anchors, every generation stored with model, decoding configuration, token usage, and timestamp—that supports reuse without regeneration.
2. Evidence that Claude’s fingerprint is *structural and rhythmic* (sentence-length burstiness, content-word density, em-dashes), survives both markdown stripping and length control, and *replicates with consistent sign across all four corpora* despite different prompt distributions and contrast sources.
3. Identification of a verbatim *interactional* signature—an offer-to-continue closer—via log-odds and suffix-automaton mining, present in Claude and absent from all other sources in our corpus.

4. A reconciliation of the “delve” folklore: the stereotyped vocabulary is a GPT/other-model trait; Claude is the cleanest frontier model on it.
5. A *coverage taxonomy* (intent $\times$ interaction structure $\times$ stance) and a reproducible audit showing that standard corpora occupy a narrow slice (single-turn, cooperative, informational). We then *act on* the audit, broadening coverage with two diverse sources—real in-the-wild prompts and ten professionally-authored task types—and show the fingerprint persists, while naming the cells that remain under-observed (multi-turn corrective, contested-stance) as targets for future work. We release the audit tooling.

2 Related Work

Detecting machine-generated text. A large literature aims to decide *whether* a text is machine-generated. Supervised detectors fine-tune classifiers such as RoBERTa on model outputs (Solaiman et al., 2019), while zero-shot methods exploit statistical signatures of the generating distribution: probability curvature (Mitchell et al., 2023), its efficient conditional-sampling successor (Bao et al., 2024), structured search over weaker-model features (Verma et al., 2024), or the cross-perplexity of a model pair (Hans et al., 2024). “In-the-wild” benchmarks (Li et al., 2024b; Wang et al., 2024) and Dugan et al. (2024) show such detectors are brittle across domains and attacks; paraphrasing in particular evades them (Krishna et al., 2023), with adversarial training (Hu et al., 2023) as a partial defense and arguments that reliable detection may be fundamentally hard (Sadasivan et al., 2023). Watermarking (Kirchenbauer et al., 2023) sidesteps detection by injecting a signal at generation time. We do not detect machine text; we instead *characterize* the style of one frontier model, assuming provenance is known.

Attributing text to a model. Closer to our aim is work on *which* model wrote a text. Surveys frame this as authorship attribution in the LLM era (Huang et al., 2024), and a recent forensic survey organizes the field into detection, attribution, and *characterization* (Kumarage et al., 2024)—the last being our category. Attribution methods range from multi-class benchmarks (Uchendu et al., 2021) to contrastive-perplexity origin tracing (Li et al., 2023a), information-density features

(Venkatraman et al., 2024), learned style representations (Rivera Soto et al., 2024), and interpretable stylometry (Kumarage and Liu, 2023); notably, stylistic signatures persist even in text optimized to fool detectors (Rivera Soto et al., 2025). We complement attribution: rather than optimizing a label, we use an interpretable, length-controlled regression and log-odds n -gram selection (Monroe et al., 2008) to surface *what* distinguishes Claude, and we audit which stylistic claims hold up under corpus coverage.

Characterizing model writing style. Most directly related is recent work that compares the *style* of LLM and human text rather than detecting it. Reinhart et al. (2025) build parallel human/LLM corpora and apply Biber’s grammatical/rhetorical features, finding systematic, scale-growing differences; Sun et al. (2025) show models are mutually distinguishable from word-level idiosyncrasies that survive paraphrase; and McGovern et al. (2024) recover model “fingerprints” from simple lexical/POS frequencies. Classic stylometry has likewise been applied to AI creative writing (O’Sullivan, 2025), and a parallel line operationalizes low-quality “AI slop” as measurable dimensions (Shaib et al., 2025). We share the parallel-corpus, interpretable-feature stance but go deeper on a single model: we characterize Claude specifically, isolate an *interactional* closing move that feature inventories do not capture, add suffix-automaton novelty evidence, and show the fingerprint replicates across four corpora.

Excess vocabulary and “AI-ese.” Kobak et al. (2025) quantify LLM-marker words (e.g. “delve”) as excess vocabulary in the biomedical literature, and corpus-scale estimators track the rising LLM footprint in papers, peer reviews, and society at large (Liang et al., 2024b,a), with parallel evidence from arXiv abstracts (Geng and Trotta, 2024) and even spoken language (Yakura et al., 2024). Popular accounts trace the habit to RLHF annotation pipelines (Willison, 2024), a hypothesis given peer-reviewed support by Juzek and Ward (2025). Such word-lists, however, are largely a ChatGPT/GPT signature and are unstable over time—“delve” fell once it was publicly flagged (Geng and Trotta, 2025). We therefore treat the “delve” folklore as a hypothesis to test rather than a given: our markers are discovered data-drivenly from parallel corpora, and we show this trait is *not* characteristic of Claude.

Stylometry and n -gram novelty. Our features build on classical stylometry, notably function-word frequencies and Burrows’s Delta (Burrows, 2002). To probe memorization (Carlini et al., 2023) versus genuine novelty, we adopt suffix-automaton n -gram search (Merrill et al., 2024a,b), unbounded n -gram indexing (Liu et al., 2024), and related corpus-search tooling (Elazar et al., 2024), applying novelty mining to a single model’s outputs comparatively. We read novelty as relative distinctiveness, not creativity: it correlates with but does not equal creative quality (Lu et al., 2024; Saakyan et al., 2026).

Post-training, persona, and behavioral signatures. The voice we characterize is a product of post-training: instruction tuning and RLHF (Ouyang et al., 2022) and, for Claude, RL from AI feedback (Bai et al., 2022) produce the “assistant” and its register, which can be framed as a trained persona (Shanahan et al., 2023) with steerable, training-induced trait directions (Chen et al., 2025); Anthropic describes Claude’s voice as deliberately shaped by character training (Anthropic, 2024). Several of our specific markers have known post-training causes: reward correlates with length (Singhal et al., 2023), preference models favor lists/bold/emphasis (Zhang et al., 2024), and RLHF reduces output diversity (Kirk et al., 2024; Padmakumar and He, 2024)—a verbosity-and-formatting-and-homogenization profile consistent with what we observe. Frontier models also exhibit behavioral signatures: verbosity bias in LLM judges (Zheng et al., 2023), sycophancy and agreement under pressure (Perez et al., 2022; Sharma et al., 2023; Xie et al., 2024), exaggerated safety/over-refusal (Röttger et al., 2024), persona/identity drift over turns (Li et al., 2024a; Choi et al., 2024), and self-interaction attractors such as the “spiritual bliss” state reported for Claude (Anthropic, 2025). These motivate our future-work agenda on multi-turn corrective behavior, where stylistic fingerprints may interact with concession and drift dynamics.

Data. We draw on parallel human/LLM corpora including HC3 (Guo et al., 2023) and AlpacaEval (Li et al., 2023b), and broaden coverage with real in-the-wild user prompts from WildChat (Zhao et al., 2024) and the professionally hand-authored No Robots dataset (Rajani et al., 2023), repurposing all of them for comparative stylometric characterization rather than detection or preference rank-

ing. Other large conversation corpora—LMSYS-Chat-1M and Chatbot Arena (Zheng et al., 2024, 2023)—offer naturally parallel, multi-model and multi-turn material for future extensions.

3 Corpus

We construct four parallel tracks (Table 1). In each, a fixed set of prompts is answered by multiple sources, so that any feature separating Claude reflects *how* it writes rather than *what* it was asked. Claude (and GPT-4o on the HC3 track) are generated by us with adaptive thinking at `effort=high`; all other cells are reused from the source datasets. Every record stores full provenance—model and version, decoding configuration, token usage, timestamp, and code commit—plus the verbatim API response.

The first two tracks (HC3, AlpacaEval) supply a human anchor and a modern multi-model contrast but, as our coverage audit shows (§3.2), they occupy a narrow behavioral slice: short, single-turn, information-seeking *questions*. The last two tracks are added precisely to broaden that slice. **WildChat** contributes real, in-the-wild first-turn user prompts (heavy on creative writing, code, and roleplay) with the original GPT-4-0314 reply as a reused contrast. **No Robots** contributes professionally hand-written prompts *and* answers, balanced across ten task types (generation, brainstorming, rewriting, summarization, classification, extraction, coding, chat, and open/closed QA); it is our second human anchor and the first one on non-QA tasks. Across all tracks there are eight distinct non-Claude models (GPT-3.5-era ChatGPT, GPT-4o, GPT-4-turbo, GPT-4-0314, Gemini-pro, DeepSeek-67B, Qwen2-72B, Llama-3-70B) and two human anchors (informal web text in HC3, professional annotators in No Robots).

3.1 Corpus statistics

Table 2 summarizes volume and length. The four tracks span a wide range of registers and lengths. Prompts grow from terse HC3/AlpacaEval questions (median 10–18 words) to substantial WildChat and No Robots tasks (median 35 and 49 words). Response lengths likewise spread: Claude’s median answer ranges from 162 words (No Robots’ often-constrained tasks) to 334 on WildChat, where its creative-writing responses push the mean to 611; the two human anchors are short and variable (HC3 median 77, No Robots

Track	Prompts	Sources (same prompt each)
HC3	200	human, ChatGPT (GPT-3.5 era), GPT-4o [†] , Claude [†]
AlpacaEval	200	gpt-4-turbo, gpt-4o, gemini-pro, deepseek-67b, Qwen2-72B, Llama-3-70B, Claude [†]
WildChat	1500	GPT-4-0314, Claude ^{†‡}
No Robots	1500	human (professional), Claude [†]

Table 1: The four parallel corpus tracks. [†] = self-generated; all other cells reused. HC3 and No Robots carry human anchors; AlpacaEval carries the modern multi-model contrast; WildChat carries real in-the-wild prompts. [‡] Claude on WildChat is currently a 628/1500 subset (generation interrupted); all WildChat comparisons use the matched 628-prompt subset, and the remaining cells will be filled and the tables regenerated.

Track	Prompts	Prompt wds (md)	Claude wds (md)	Total words
HC3	200	10	267	156k
AlpacaEval	200	18	230	354k
WildChat	1500 [‡]	35	334	946k
No Robots	1500	49	162	397k

Table 2: Corpus volume and length (median word counts). Totals sum all sources in the track. [‡] Claude answered 628 of the 1500 WildChat prompts so far. Full per-source distributions: Appendix B.

median 48). We account for length via length controls (§5.3). Per-domain volume and full distributions are in Appendix B. A corpus-level register signal already foreshadows the results: Claude’s responses are the most *interrogative* at the sentence level on every track (17%/10%/6%/9% of sentences on HC3/AlpacaEval/WildChat/No Robots, vs. $\leq 5\%$ for every contrast source—the offer-closer, §5.5), and Claude is denser than its human anchors (Flesch ≈ 40 –48).

3.2 Coverage: auditing and expanding the behavioral slice

Because style is context-dependent, we audit *what behavior the prompts elicit*, labeling every prompt by intent, speech act, register, tone, and topic, plus interaction flags (src/analyze_coverage.py; large tracks sampled at $n=400$). Labels are produced by Claude Opus 4.8 itself (the model under study), a circularity we flag in Limitations.

TODO (human validation pending): hand-annotate a ~ 50 -prompt sample, report inter-annotator agreement with the LLM labels, and add a small validation table. Quick human pass to be added.

The audit shows that HC3 and AlpacaEval alone occupy a narrow region: HC3 is 96% explanation/QA/advice and 75% *questions*; AlpacaEval is broader but still question/directive and information-oriented. Treating this as a coverage problem to fix rather than merely report, we add the WildChat and No Robots tracks, and the four tracks are *complementary* (Table 3): WildChat is dominated by creative writing (42%) and code (11%), No Robots by classification/ extraction (25%), summarization/rewriting (19%), and QA (20%). Intents that were absent or near-absent in the first two tracks—creative writing, code, role-play, summarization, rewriting, classification, extraction, and personal/emotional—are now substantially represented. The expansion also shifts the other axes (Appendix B): the dominant speech act flips from HC3’s 75% *questions* to 59–73% *directives* on the new tracks; a non-trivial *formal* register appears (10% of WildChat prompts) where it was nearly absent; and topics broaden from HC3’s health/science/finance concentration to include entertainment, software, the arts, and personal life.

Two limits remain, and we are explicit about them. First, the corpus is still *single-turn*: only WildChat shows a non-trivial fraction of prompts that imply follow-up (7%), and essentially none are adversarial or corrective, so behaviors gated on a multi-turn *declarative* challenge—e.g. the concession move “you’re right to push back on that”—remain out of scope and motivate the multi-turn extension (§5.9). Second, contested-stance prompts (those that hand Claude an assertion to agree with or contest) are still a small minority on every track. The fingerprint we report should therefore be read as Claude’s *single-turn* voice—but now across a markedly broader range of single-turn tasks than informational QA alone.

4 Methods

We compute interpretable features per response (lexical: type-token ratio, hapax rate, function-word rate; punctuation: em-dash, colon, question, exclamation, emoji; syntax: mean sentence length and *burstiness*, the ratio of the standard deviation to the mean of words-per-sentence; mark-down/structure; and rhetoric). Rates are per 100 or 1000 words so that text length does not mechanically dominate. We compute two dozen such features per response. We summarize source differences with Cohen’s d (Claude vs. a compari-

Intent (primary, %)	HC3	Alpaca	WildC	NoRob
explanation	47	17	9	4
factual QA	27	15	11	20
advice/recommendation	25	16	5	9
instruction task	2	13	8	2
creative writing	0	12	42	10
code	0	6	11	10
summarize/rewrite/edit	0	4	3	19
classification/extraction	0	5	1	25
opinion/subjective	0	6	2	0
math/roleplay/other	1	11	9	2
<i>flag</i> : multi-turn impl.	4	2	7	1
<i>flag</i> : subjective	7	20	11	8
<i>flag</i> : sensitive	31	8	20	6

Table 3: Prompt intent distribution and interaction flags, % of audited prompts (HC3/AlpacaEval $n=200$; WildChat/No Robots sampled $n=400$; `coverage_labels.csv`). The four tracks are complementary: bold marks each new track’s distinctive contribution. Multi-turn and contested-stance prompts remain scarce on all tracks.

son group); all reported effects are significant at $p < 0.001$ (Welch’s t , $n=200$ Claude vs. ≥ 200 comparison). The classifier is ℓ_2 -regularized logistic regression ($C=1.0$, balanced class weights). To separate *voice* from *formatting*, we additionally compute all prose features after stripping markdown markup. We mine characteristic n -grams with the Monroe et al. (2008) log-odds estimator; quantify length-independence with OLS (§5.3); measure separability with ℓ_2 -regularized logistic regression under 5-fold cross-validation; and mine arbitrary-length spans and n -gram novelty with a suffix automaton (Merrill et al., 2024a).

5 Results

5.1 A large signal, dominated at first by formatting

The strongest raw separators between Claude and the other HC3 sources are markdown-structural: markdown headers ($d=2.9$), bold ($d=2.3$), and bullets ($d=1.8$), which the human and old-ChatGPT cells use essentially never. This is real but is a *formatting* habit; the decisive question is whether anything survives removing it.

5.2 The voice survives markdown stripping

With all markup stripped, Claude still separates strongly on prose (Table 4): it interleaves short and long sentences far more than any other source (burstiness), writes denser content-word prose (lower function-word rate), and retains a marked

Prose feature	$ d $	Claude / others
Sentence burstiness	1.69	0.86 / 0.36–0.55
Function-word rate ↓	1.47	0.33 / 0.37–0.44
Em-dash rate	0.87	0.88 / ≤ 0.40
Colon rate	0.84	1.41 / 0.24–1.69
Question rate	0.82	0.61 / 0.01–0.28

Table 4: HC3 track, Claude vs. pooled others, markdown stripped. Mean |Cohen’s d | and group means (Claude / range over the other sources). ↓ = Claude lower.

em-dash habit. These are the load-bearing stylistic findings.

5.3 The signature is not a length artifact

On the HC3 track, Claude writes longer answers (mean 253 words vs. 120 for humans, 228 for GPT-4o), so we test whether the prose effects merely track length. Within every word-count quartile the effects persist, and a length-controlled regression (HC3) (feature $\sim$ is_claude + $z(n_words)$) leaves the Claude coefficient large and highly significant: burstiness $t=20.6$, function-word rate $t=-14.4$, em-dash $t=12.6$, question rate $t=4.6$. Type-token ratio is length-sensitive and we treat lexical-diversity claims as length-conditioned. The effects also hold *within every domain* (Appendix C): across all nine domains of the HC3 and AlpacaEval tracks, burstiness has $d > 0$, function-word rate $d < 0$ (Claude denser), and em-dash and question rate $d > 0$, so the fingerprint is not a topic artifact; the next section extends this check to two further corpora.

5.4 The fingerprint replicates across all four tracks

The prose and length analyses above are on HC3. Do the same features distinguish Claude when the prompt distribution and the contrast source change? We compute Claude-vs-pooled Cohen’s d separately within each track (for WildChat, on the matched 628-prompt subset, preserving the parallel-corpus control; `src/analyze_tracks.py`). The headline features carry the *same sign in all four tracks* (Table 5): Claude writes with higher sentence-length burstiness, denser content-word prose (lower function-word rate), more em-dashes, more markdown scaffolding, and more questions—against a human panel, against six modern models, against GPT-4o on real prompts, and against professional human writers on diverse tasks. Magni-

Feature	HC3	Alp	WiC	NoR	sign
sentence burstiness	1.56	0.45	0.58	0.63	+4/4
function-word rate	-1.29	-0.57	-0.49	-0.24	-4/4
em-dash rate	0.86	0.12	0.48	0.34	+4/4
markdown headers	4.06	1.81	1.26	0.66	+4/4
markdown bold	2.72	0.88	1.12	1.09	+4/4
question rate	0.35	0.40	0.32	0.03	+4/4

Table 5: Claude-vs-pooled Cohen’s d within each track (raw features). The sign is consistent across all four tracks for every headline feature; magnitude is largest where the contrast is cleanest (HC3). The question/offer habit is robust except on No Robots, whose task mix does not invite continuation.

tudes are largest on HC3, whose short, informal contrast cells make the gap widest, and shrink on the harder, more varied contrasts, but the direction never flips. (Two weaker features— colon rate and type-token ratio—are *not* sign-stable across tracks and we do not claim them as part of the fingerprint.) The interactional question habit is the one headline feature that fades on No Robots ($d=0.03$), whose classification, extraction, and rewriting tasks do not invite an offer to continue—a sensible boundary condition rather than a failure (§5.5).

5.5 The interactional signature: an offer-to-continue closer

Data-driven n -gram mining (Monroe et al., 2008) surfaces a clear interactional fingerprint rather than a content-word list. The most Claude-like tri-grams (HC3) are *would you like*, *like me to*, *me to explain*, *in more detail*, *to go deeper*; the anti-signature—terms Claude *underuses*—is led by the formal frame *it is important to*. At the sentence level this closer registers as a striking jump in interrogatives: **17%/10% of Claude’s response sentences are questions (HC3/AlpacaEval), versus $\leq 4\%$ for every other source** (Appendix B). Suffix-automaton mining (§5.8) confirms the pattern at arbitrary length. Figure 1 shows the fingerprint in a single matched pair.

5.6 The “delve” vocabulary is a GPT trait Claude avoids

On the AlpacaEval track—an essay/instruction genre that *does* elicit the stereotyped words—Claude is the cleanest of seven models (Table 6). It is the only model that never writes *delve* or *delves*, and it uses *crucial* least of all seven (0.039 per 1k vs. DeepSeek 0.054, Gemini 0.090, Llama 0.185,

Prompt (ELI5): *Why is skin itchy?*

Human: “As far as I know, scratching is an evolved reaction to help stop insects and other parasites from sucking our blood...”

Claude Opus 4.8: “...scratching sends a *new* signal to your brain that covers up the itchy feeling. But scratching too much can hurt your skin, so it’s better to be gentle!

Would you like me to explain anything else?” [closes with an emoji]

Figure 1: Same prompt, two sources (excerpts). Claude’s answer is longer, more structured, and ends with the offer-to-continue closer; the human answer is terse and declarative.

per-1k	Claude	g4t	g4o	qwen	llama
delve	0	.018	.018	0	.017
delves	0	.037	.089	.019	.034
crucial	.039	.422	.340	.346	.185
intricate	0	0	0	.019	.050
pivotal	0	.073	.018	.038	0

Table 6: Frequency (per 1000 words) of stereotyped “excess” words on the AlpacaEval essay track. g4t = gpt-4-turbo, g4o = gpt-4o. Claude is at or near the minimum for every term. (gemini/deepseek omitted for space; same pattern.)

GPT-4o 0.340, Qwen 0.346, GPT-4-turbo 0.422). Claude’s own lexical lean is conversational (“actually” occurs $\sim 6\times$ more often in Claude than GPT-4o on HC3); this conversational lean is a secondary signal beneath the structural one.

5.7 Separability

An interpretable classifier confirms a stable, learnable voice (Table 7). Stripping markdown reduces AUC **only** modestly, so the classifier is not merely reading layout; and its top coefficients are exactly the features of §5.2 (burstiness, function-word density, em-dash, question rate), an independent-method confirmation. Raw length is not a feature, and residualizing every feature on response length barely changes separation (prose-only AUC 0.946 $\rightarrow$ 0.931 on HC3, 0.867 $\rightarrow$ 0.861 on AlpacaEval), so the result is not a length artifact.

5.8 Arbitrary-length spans and n -gram novelty

Suffix-automaton mining (Merrill et al., 2024a) gives the most concrete form of the signature. Within each track, spans up to *eight words* occur dozens of times in Claude and *zero* times across that track’s non-Claude sources (HC3: humans + 2 models; AlpacaEval: 6 models, no human): on

Track (contrast)	w/ fmt	prose
HC3 (human, GPT-4o, 3.5)	0.982	0.946
AlpacaEval (6 modern)	0.901	0.867
WildChat (GPT-4-0314)	0.973	0.854
No Robots (human, pro)	0.922	0.788

Table 7: Claude-vs-rest ROC-AUC (logistic regression, 5-fold CV) on all four tracks. Claude is highly separable under every contrast, including from prose alone; the lowest prose-only AUC (No Robots, 0.79) is the hardest case—against professional human writers on short, constrained tasks—and is still well above chance.

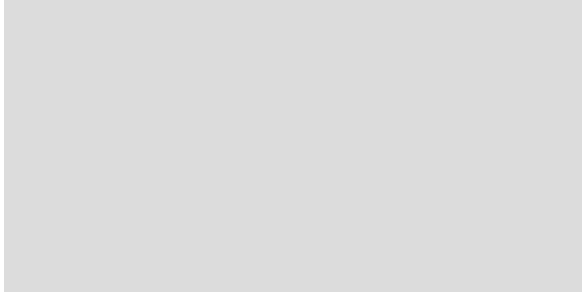


Figure 2: **[Placeholder image—replace with real plot.]** n -gram novelty by source: the fraction of each source’s n -grams ($n=1\dots 10$) that are absent from the pooled other sources (AlpacaEval track). Claude’s curve sits above all six other models at every n (bigram novelty 0.519 vs. 0.27–0.36), i.e. its phrasing overlaps least with the pool. Data: reports/ngram_dawg.md.

Replace the grey placeholder with the real novelty-curve figure (one line per source).

HC3, *would you like me to explain any* ($\times 20$) and *would you like me to go deeper into* ($\times 14$); on AlpacaEval, *would you like me to expand on any* ($\times 7$). Claude also has the highest n -gram novelty of any source (Figure 2; e.g. bigram novelty 0.519 vs. 0.27–0.36 for the six AlpacaEval models): its exact phrasing overlaps least with the pool. We read novelty as relative distinctiveness, not absolute creativity (see Limitations).

5.9 Secondary analyses (in progress)

Three planned analyses are not yet included; designs and placeholder tables appear in Appendix D.

- **Effect of reasoning effort/thinking.** Whether the fingerprint intensifies or attenuates across $\text{effort} \in \{\text{low, medium, high}\}$ and with thinking disabled. [To be completed; Appendix D.1.]

- **Across Claude versions.** Intra-Claude drift across model versions (e.g. Opus/Sonnet/Haiku and older releases). [To be completed; Appendix D.2.]
- **Multi-turn and self-interaction.** Turn-dependent traits and the Claude-to-Claude “spiritual bliss” attractor reported in (Anthropic, 2025). [To be completed; Appendix D.3.]

6 Discussion

Across four corpora, five methods, eight contrast models, and two human anchors, the same fingerprint recurs and survives every control: Claude-speak is markdown-scaffolded, rhythmically varied, content-dense, and engagement-and-offer-oriented prose—and specifically *not* the “delve” register. That the signature holds in sign from terse factual QA to in-the-wild creative writing to constrained extraction tasks suggests it is a property of the model rather than of any one prompt genre. A methodological lesson follows: borrowed AI-word lists largely measure *GPT*, and characterizing a specific model requires data-driven discovery against topic-matched references.

Limitations

We study a single Claude configuration (Opus 4.8, $\text{effort}=\text{high}$); the effort, version, and multi-turn axes are not yet measured (§5.9). Our coverage expansion (§3.2) broadens the corpus well beyond informational QA—into creative writing, code, summarization, rewriting, classification, and extraction—but it remains *single-turn*: context-gated behaviors that require a multi-turn declarative challenge (corrective concession, sustained refusal) are still out of scope and are the target of the planned multi-turn extension (English only). The WildChat Claude cells are currently a 628/1500 subset (generation interrupted); all WildChat comparisons use the matched subset, and the remaining cells and refreshed tables are pending. Reused contrast completions are their published vintages (e.g. GPT-4-0314, gemini-pro 1.0, Qwen2-72B, GPT-4o-turbo/4o 2024), modern-ish but not the absolute latest. n -gram novelty is measured relative to our corpus pool, not a large external reference, and may be inflated for Claude as the lone out-of-cluster (and only freshly generated) source. The two human anchors differ in register—HC3 is informal web text, No Robots is professional

annotation—which lets us check that the human–Claude gap is not merely an artifact of informal writing (the fingerprint holds against both), though neither is edited published prose. The coverage audit’s labels are produced by an LLM (the model under study), introducing a circularity; a human-validation pass on a labeled sample is pending. Finally, raw length is excluded from the classifier features and the feature effects are length-controlled (§5.3), but some prose features still correlate with length; we report a length-residualized classifier as a check (§5.7).

Data and Reproducibility

We do not test mechanism, but the interactional, low-formality, offer-oriented profile is consistent with post-training (RLHF) that rewards helpful, engaging, and appropriately hedged responses. Code and the fully provenanced corpus (generations plus metadata, and the audit tooling) will be released at an anonymized URL. HC3, AlpacaEval, WildChat, and No Robots are public research artifacts used under their licenses; we redistribute only derived statistics and our own model generations.

Ethics Statement

This work characterizes model writing style using public datasets and model outputs; it involves no human subjects or private data. Stylistic fingerprints can aid provenance and detection but could also inform evasion; we report aggregate, interpretable features rather than a deployable detector.

References

- Anthropic. 2024. Claude’s character. <https://www.anthropic.com/research/claude-character>.
- Anthropic. 2025. *System card: Claude opus 4 & claude sonnet 4*. Technical report, Anthropic.
- Yuntao Bai, Saurav Kadavath, Sandipan Kundu, Amanda Askell, Jackson Kernion, Andy Jones, Anna Chen, and 1 others. 2022. Constitutional AI: Harmlessness from AI feedback. ArXiv:2212.08073.
- Guangsheng Bao, Yanbin Zhao, Zhiyang Teng, Linyi Yang, and Yue Zhang. 2024. Fast-DetectGPT: Efficient zero-shot detection of machine-generated text via conditional probability curvatures. In *International Conference on Learning Representations (ICLR)*. ArXiv:2310.05130.

- John Burrows. 2002. ‘delta’: a measure of stylistic difference and a guide to likely authorship. *Literary and Linguistic Computing*, 17(3):267–287.
- Nicholas Carlini, Daphne Ippolito, Matthew Jagielski, Katherine Lee, Florian Tramèr, and Chiyuan Zhang. 2023. Quantifying memorization across neural language models. In *International Conference on Learning Representations (ICLR)*. ArXiv:2202.07646.
- Runjin Chen, Andy Ardit, Henry Sleight, Owain Evans, and Jack Lindsey. 2025. Persona vectors: Monitoring and controlling character traits in language models. ArXiv:2507.21509.
- Junhyuk Choi, Yeseon Hong, Minju Kim, and Bugeun Kim. 2024. Examining identity drift in conversations of LLM agents. ArXiv:2412.00804.
- Liam Dugan, Alyssa Hwang, Filip Trhlik, Josh Magnus Ludan, Andrew Zhu, Hainiu Xu, Daphne Ippolito, and Chris Callison-Burch. 2024. RAID: A shared benchmark for robust evaluation of machine-generated text detectors. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*. ArXiv:2405.07940.
- Yanai Elazar, Akshita Bhagia, Ian Magnusson, Abhिलाषा Ravichander, Dustin Schwenk, Alane Suhr, Pete Walsh, Dirk Groeneveld, Luca Soldaini, Sameer Singh, Hannaneh Hajishirzi, Noah A. Smith, and Jesse Dodge. 2024. What’s in my big data? In *International Conference on Learning Representations (ICLR)*. ArXiv:2310.20707.
- Mingmeng Geng and Roberto Trotta. 2024. Is ChatGPT transforming academics’ writing style? ArXiv:2404.08627.
- Mingmeng Geng and Roberto Trotta. 2025. Human-LLM coevolution: Evidence from academic writing. ArXiv:2502.09606.
- Biyang Guo, Xin Zhang, Ziyuan Wang, Minqi Jiang, Jinran Nie, Yuxuan Ding, Jianwei Yue, and Yupeng Wu. 2023. How close is ChatGPT to human experts? comparison corpus, evaluation, and detection. *arXiv preprint arXiv:2301.07597*.
- Abhimanyu Hans, Avi Schwarzschild, Valeriia Cherepanova, Hamid Kazemi, Aniruddha Saha, Micah Goldblum, Jonas Geiping, and Tom Goldstein. 2024. Spotting LLMs with binoculars: Zero-shot detection of machine-generated text. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*. ArXiv:2401.12070.
- Xiaomeng Hu, Pin-Yu Chen, and Tsung-Yi Ho. 2023. RADAR: Robust AI-text detection via adversarial learning. In *Advances in Neural Information Processing Systems (NeurIPS)*. ArXiv:2307.03838.
- Baixiang Huang, Canyu Chen, and Kai Shu. 2024. Authorship attribution in the era of LLMs: Problems, methodologies, and challenges. *ACM SIGKDD Explorations Newsletter*. ArXiv:2408.08946.

654	Tom S. Juzek and Zina B. Ward. 2025. Why does	Weixin Liang, Zachary Izzo, Yaohui Zhang, and 1 oth-	709
655	ChatGPT “delve” so much? exploring the sources	ers. 2024a. Monitoring AI-modified content at scale:	710
656	of lexical overrepresentation in large language mod-	A case study on the impact of ChatGPT on AI con-	711
657	els. In <i>Proceedings of the 31st International Con-</i>	ference peer reviews. In <i>International Conference</i>	712
658	<i>ference on Computational Linguistics (COLING)</i> .	<i>on Machine Learning (ICML)</i> . ArXiv:2403.07183.	713
659	ArXiv:2412.11385.		
660	John Kirchenbauer, Jonas Geiping, Yuxin Wen,	Weixin Liang, Yaohui Zhang, Zhengxuan Wu, Haley	714
661	Jonathan Katz, Ian Miers, and Tom Goldstein. 2023.	Lepp, Wenlong Ji, Xuandong Zhao, Hancheng Cao,	715
662	A watermark for large language models. In <i>Proceeed-</i>	Sheng Liu, Siyu He, Zhi Huang, Diyi Yang, Christo-	716
663	<i>ings of the 40th International Conference on Ma-</i>	pher Potts, Christopher D. Manning, and James Y.	717
664	<i>chine Learning (ICML)</i> . ArXiv:2301.10226.	Zou. 2024b. Mapping the increasing use of LLMs in	718
		scientific papers. <i>arXiv preprint arXiv:2404.01268</i> .	719
665	Robert Kirk, Ishita Mediratta, Christoforos Nalmpantis,	Jiacheng Liu, Sewon Min, Luke Zettlemoyer, Yejin	720
666	Jelena Luketina, Eric Hambro, Edward Grefenstette,	Choi, and Hannaneh Hajishirzi. 2024. Infini-	721
667	and Roberta Raileanu. 2024. Understanding the ef-	gram: Scaling unbounded n -gram language mod-	722
668	fects of RLHF on LLM generalisation and diversity.	els to a trillion tokens. In <i>Proceedings of the</i>	723
669	In <i>International Conference on Learning Representa-</i>	<i>First Conference on Language Modeling (COLM)</i> .	724
670	<i>tions (ICLR)</i> . ArXiv:2310.06452.	ArXiv:2401.17377.	725
671	Dmitry Kobak, Rita González-Márquez, Emőke-Ágnes	Ximing Lu, Melanie Sclar, Skyler Hallinan, Niloofar	726
672	Horvát, and Jan Lause. 2025. Delving into LLM-	Mirehghallah, Jiacheng Liu, Seungju Han, Allyson	727
673	assisted writing in biomedical publications through	Ettinger, Liwei Jiang, Khyathi Chandu, Nouha Dziri,	728
674	excess vocabulary. <i>Science Advances</i> , 11(27).	and Yejin Choi. 2024. AI as humanity’s salieri:	729
675	ArXiv:2406.07016.	Quantifying linguistic creativity of language mod-	730
		els via systematic attribution of machine text against	731
676	Kalpesh Krishna, Yixiao Song, Marzena Karpinska,	web text. ArXiv:2410.04265.	732
677	John Wieting, and Mohit Iyyer. 2023. Paraphras-		
678	ing evades detectors of AI-generated text, but re-	Hope McGovern, Rickard Stureborg, Yoshi Suhara,	733
679	trieval is an effective defense. In <i>Advances in</i>	and Dimitris Alikaniotis. 2024. Your large language	734
680	<i>Neural Information Processing Systems (NeurIPS)</i> .	models are leaving fingerprints. ArXiv:2405.14057.	735
681	ArXiv:2303.13408.		
682	Tharindu Kumarage, Garima Agrawal, Paras Sheth,	William Merrill, Noah A. Smith, and Yanai Elazar.	736
683	Raha Moraffah, Aman Chadha, Joshua Garland, and	2024a. Evaluating n -gram novelty of language mod-	737
684	Huan Liu. 2024. A survey of AI-generated text	els using Rusty-DAWG. In <i>Proceedings of the 2024</i>	738
685	forensic systems: Detection, attribution, and charac-	<i>Conference on Empirical Methods in Natural Lan-</i>	739
686	terization. ArXiv:2403.01152.	<i>guage Processing (EMNLP)</i> . ArXiv:2406.13069.	740
687	Tharindu Kumarage and Huan Liu. 2023. Neural au-	William Merrill, Noah A. Smith, and Yanai Elazar.	741
688	thorship attribution: Stylometric analysis on large	2024b. Evaluating n -gram novelty of language mod-	742
689	language models. ArXiv:2308.07305.	els using rusty-DAWG. In <i>Proceedings of EMNLP</i> .	743
		ArXiv:2406.13069.	744
690	Kenneth Li, Tianle Liu, Naomi Bashkansky, David	Eric Mitchell, Yoonho Lee, Alexander Khazatsky,	745
691	Bau, Fernanda Viégas, Hanspeter Pfister, and Mar-	Christopher D. Manning, and Chelsea Finn. 2023.	746
692	tin Wattenberg. 2024a. Measuring and controlling	DetectGPT: Zero-shot machine-generated text detec-	747
693	instruction (in)stability in language model dialogs.	tion using probability curvature. In <i>Proceedings</i>	748
694	ArXiv:2402.10962.	<i>of the 40th International Conference on Machine</i>	749
		<i>Learning (ICML)</i> . ArXiv:2301.11305.	750
695	Linyang Li, Pengyu Wang, Ke Ren, Tianxiang Sun, and	Burt L. Monroe, Michael P. Colaresi, and Kevin M.	751
696	Xipeng Qiu. 2023a. Origin tracing and detecting of	Quinn. 2008. Fightin’ words: Lexical feature se-	752
697	LLMs. ArXiv:2304.14072.	lection and evaluation for identifying the content of	753
		political conflict. <i>Political Analysis</i> , 16(4):372–403.	754
698	Xuechen Li, Tianyi Zhang, Yann Dubois, Rohan	James O’Sullivan. 2025. Stylometric comparisons of	755
699	Taori, Ishaan Gulrajani, Carlos Guestrin, Percy	human versus AI-generated creative writing. <i>Hu-</i>	756
700	Liang, and Tatsunori B. Hashimoto. 2023b. Al-	<i>manities and Social Sciences Communications</i> .	757
701	pacaEval: An automatic evaluator of instruction-		
702	following models. https://github.com/tatsu-lab/alpaca_eval .	Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida,	
		Carroll L. Wainwright, Pamela Mishkin, Chong	
		Zhang, Sandhini Agarwal, Katarina Slama, Alex	
		Ray, and 1 others. 2022. Training language mod-	
		els to follow instructions with human feedback. In	
		<i>Advances in Neural Information Processing Systems</i>	
		<i>(NeurIPS)</i> . ArXiv:2203.02155.	
	Yafu Li, Qintong Li, Leyang Cui, Wei Bi, Zhilin		
	Wang, Longyue Wang, Linyi Yang, Shuming Shi,		
	and Yue Zhang. 2024b. MAGE: Machine-generated		
	text detection in the wild. In <i>Proceedings of ACL</i> .		
	ArXiv:2305.13242.		

765	Vishakh Padmakumar and He He. 2024. Does writing with language models reduce content diversity? In <i>International Conference on Learning Representations (ICLR)</i> . ArXiv:2309.05196.	820
766		821
767		822
768		
769	Ethan Perez, Sam Ringer, Kamilė Lukošiuūtė, Karina Nguyen, Edwin Chen, and 1 others. 2022. Discovering language model behaviors with model-written evaluations. <i>arXiv preprint arXiv:2212.09251</i> .	823
770		824
771		825
772		
773	Nazneen Rajani, Lewis Tunstall, Edward Beeching, Nathan Lambert, Alexander M. Rush, and Thomas Wolf. 2023. No robots. https://huggingface.co/datasets/HuggingFaceH4/no_robots .	826
774		827
775		828
776		829
777		830
		831
		832
778	Alex Reinhart, David West Brown, Ben Markey, Michael Laudenbach, Kachata Pantusen, Ronald Yurko, and Gordon Weinberg. 2025. Do LLMs write like humans? variation in grammatical and rhetorical styles. <i>Proceedings of the National Academy of Sciences</i> , 122(8). ArXiv:2410.16107.	833
779		834
780		835
781		836
782		
783		
784	Rafael Rivera Soto, Barry Chen, and Nicholas Andrews. 2025. Attacks on machine-text detectors retain stylistic fingerprints. ArXiv:2505.14608.	837
785		838
786		839
		840
		841
787	Rafael Rivera Soto, Kailin Koch, Aleem Khan, Barry Chen, Marcus Bishop, and Nicholas Andrews. 2024. Few-shot detection of machine-generated text using style representations. In <i>International Conference on Learning Representations (ICLR)</i> . ArXiv:2401.06712.	842
788		843
789		844
790		845
791		
792		
793	Paul Röttger, Hannah Rose Kirk, Bertie Vidgen, Giuseppe Attanasio, Federico Bianchi, and Dirk Hovy. 2024. XSTest: A test suite for identifying exaggerated safety behaviours in large language models. In <i>Proceedings of NAACL</i> . ArXiv:2308.01263.	846
794		847
795		848
796		849
797		
798	Arkadiy Saakyan, Najoung Kim, Smaranda Muresan, and Tuhin Chakrabarty. 2026. Death of the novel(ty): Beyond n -gram novelty as a metric for textual creativity. In <i>International Conference on Learning Representations (ICLR)</i> . ArXiv:2509.22641.	850
799		851
800		852
801		853
802		854
803		855
		856
804	Vinu Sankar Sadasivan, Aounon Kumar, Sriram Balasubramanian, Wenxiao Wang, and Soheil Feizi. 2023. Can AI-generated text be reliably detected? ArXiv:2303.11156.	857
805		858
806		859
807		860
808	Chantal Shaib, Tuhin Chakrabarty, Diego Garcia-Olano, and Byron C. Wallace. 2025. Measuring AI “slop” in text. ArXiv:2509.19163.	861
809		862
810		863
		864
811	Murray Shanahan, Kyle McDonell, and Laria Reynolds. 2023. Role play with large language models. <i>Nature</i> , 623:493–498. ArXiv:2305.16367.	865
812		866
813		867
		868
		869
		870
		871
		872
		873
		874
		875
		876
		877
		878
		879
		880
		881
		882
		883
		884
		885
		886
		887
		888
		889
		890
		891
		892
		893
		894
		895
		896
		897
		898
		899
		900
		901
		902
		903
		904
		905
		906
		907
		908
		909
		910
		911
		912
		913
		914
		915
		916
		917
		918
		919
		920
		921
		922
		923
		924
		925
		926
		927
		928
		929
		930
		931
		932
		933
		934
		935
		936
		937
		938
		939
		940
		941
		942
		943
		944
		945
		946
		947
		948
		949
		950
		951
		952
		953
		954
		955
		956
		957
		958
		959
		960
		961
		962
		963
		964
		965
		966
		967
		968
		969
		970
		971
		972
		973
		974
		975
		976
		977
		978
		979
		980
		981
		982
		983
		984
		985
		986
		987
		988
		989
		990
		991
		992
		993
		994
		995
		996
		997
		998
		999
		1000

Wenting Zhao, Xiang Ren, Jack Hessel, Claire Cardie, Yejin Choi, and Yuntian Deng. 2024. WildChat: 1M chatGPT interaction logs in the wild. In *International Conference on Learning Representations (ICLR)*.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Tianle Li, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zhuohan Li, Zi Lin, Eric P. Xing, Joseph E. Gonzalez, Ion Stoica, and Hao Zhang. 2024. LMSYS-Chat-1M: A large-scale real-world LLM conversation dataset. ArXiv:2309.11998.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. Judging LLM-as-a-judge with MT-bench and chatbot arena. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks*. ArXiv:2306.05685.

A Feature definitions

Full definitions of all interpretable features (rates, burstiness, markdown and rhetorical markers) are provided with the released code (`src/features.py`).

B Full corpus statistics

All statistics here are reproducible via `src/dataset_stats.py` (volume, lengths, sentence mood, register) and `src/analyze_coverage.py` (intent).

Response length by source (words, mean/median). HC3: human 120/77, ChatGPT 176/178, GPT-4o 228/239, Claude 254/267. AlpacaEval: deepseek-67b 186/184, gemini-pro 222/234, Claude 254/230, Qwen2-72B 260/259, gpt-4-turbo 272/285, gpt-4o 280/285, Llama-3-70B 297/308. WildChat: GPT-4-0314 375/357, Claude 611/334 (the high mean is driven by long creative-writing responses). No Robots: human 90/48, Claude 175/162.

Volume by domain / task (prompts; total response words, all sources). HC3: general 80/47k, finance 40/40k, cs_ai 40/35k, medicine 40/33k. AlpacaEval: selfinstruct 64/79k, oasst 49/96k, helpful_base 35/70k, koala 35/67k, vicuna 17/43k. No Robots (balanced, 150 prompts each): generation, brainstorm, coding, and rewrite are the highest-volume task types; classify, extract, and closed-QA the lowest (short answers). WildChat carries no source task labels (real user logs); its task mix is the LLM-labeled distribution in Table 3.

Axis	Value	HC3	Alp	WiC	NoR
speech act	question	75	53	23	38
	directive	23	47	74	59
	assertion	0	0	2	0
register	neutral	65	65	62	79
	casual	35	31	27	19
	formal	0	3	10	1
tone	neutral	82	88	86	90
	emotional	10	2	1	1
	playful	0	9	11	7

Table 8: Prompt variety (% of audited prompts) along *pragmatic* speech act (LLM-labeled), register, and tone. HC3/AlpacaEval $n=200$; WildChat/No Robots sampled $n=400$. The added tracks shift the dominant speech act to directives and introduce a formal register; non-neutral tone and assertions stay rare. Topic and intent breakdowns: `coverage_labels.csv`. (Syntactic mood, which differs, is in §3.2.)

Sentence mood (% of response sentences). Interrogative sentences: Claude 17%/10%/6%/9% (HC3/AlpacaEval/WildChat/No Robots) vs. 1–5% for every contrast source—Claude is the most interrogative on every track. Imperative and exclamatory sentences are $\leq 2\%$ and $\leq 7\%$ respectively throughout.

Register. First-/second-person pronoun rates per 100 words and Flesch Reading Ease are in `stats_register.csv`. Claude is denser (lower Flesch) than both human anchors (HC3 human 48, No Robots human 59, vs. Claude ≈ 40 –48) and than the modern-model panel on AlpacaEval; on WildChat its readability is comparable to GPT-4-0314’s.

Prompt variety (audit). Where HC3 and AlpacaEval are narrow—almost all *questions/directives*, overwhelmingly *neutral* tone, and nearly no *formal* register—the two added tracks broaden every axis (Table 8): directives become the dominant speech act, a non-trivial formal register appears on WildChat (10%), and topics spread from HC3’s health/science/finance concentration to entertainment, software, the arts, and personal life. Non-neutral tone and contested-stance assertions remain scarce on all four tracks.

Coverage expansion (delivered, and remaining). The WildChat and No Robots tracks (§3.2) populate intents that HC3/AlpacaEval lacked: creative writing, code, roleplay, summarization, rewriting, classification, and extraction. Two cells remain under-observed and motivate future work: (i)

Track	Domain	burst.	func↓	em-d.	ques.
HC3	cs/ai	2.50	−1.86	1.12	1.02
HC3	finance	2.11	−1.75	0.94	1.55
HC3	general	1.46	−0.97	0.95	0.17
HC3	medicine	2.65	−1.28	2.41	0.72
Alpaca	helpful	1.19	−0.81	0.66	0.85
Alpaca	koala	0.99	−0.43	2.02	0.25
Alpaca	oasst	1.46	−0.77	1.38	1.26
Alpaca	selfinst.	0.56	−0.40	0.26	0.29
Alpaca	vicuna	1.98	−1.17	3.22	2.21

Table 9: Per-domain Claude-vs-others Cohen’s d , prose only. burst. = sentence burstiness; func↓ = function-word rate (negative = Claude denser); em-d. = em-dash; ques. = question rate. Uniform signs across domains.

comparatively; (ii) elicited Claude-to-Claude conversations to test the “spiritual bliss” attractor (Anthropic, 2025) quantitatively (e.g. trajectory of consciousness/gratitude vocabulary and vocabulary collapse over turns). [Results table to be inserted.]

multi-turn corrective scenarios (a user challenges or corrects Claude) to elicit concession / push-back behavior, and (ii) a small, defensively-framed sensitive/refusal set. The audit tooling re-runs unchanged on any added track.

C Per-domain robustness

Claude-vs-pooled-others Cohen’s d (markdown-stripped prose) computed *within* each domain (Table 9). The signs are uniform across all nine domains—burstiness and em-dash/question positive, function-word rate negative (Claude denser)—so the fingerprint is not driven by any single topic.

D Placeholders for secondary analyses

D.1 Effect of reasoning effort

Planned design. Regenerate the 200 HC3 prompts (and 200 AlpacaEval prompts) under `effort` ∈ {low, medium, high} and with thinking disabled, holding all else fixed; recompute the §5.2 features and the §5.7 classifier within each setting; report whether burstiness, density, em-dash, and the offer-closer scale with effort. [Results table to be inserted.]

D.2 Across Claude versions

Planned design. Generate the same prompts with additional Claude versions (e.g. Sonnet/Haiku, and an older 3.x), then measure intra-Claude drift in the fingerprint features and the offer-closer rate. [Results table to be inserted.]

D.3 Multi-turn and self-interaction

Planned design. (i) Two-turn parallel probes (e.g. MT-Bench-style) to measure turn-dependent traits